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Review

Lipid bilayer technologies in ion channel recordings and their potential in drug screening assay

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ABSTRACT

Ion channels are expressed in every tissue and are important for many physiological processes. Malfunction of ion channels can lead to diseases in many tissues. All these make ion channels ideal targets for drug development. Many techniques are employed to study ion channel functions including the lipid bilayer technique. Lipid bilayer membranes are being synthesized in vitro to mimic natural cell membranes. Ion channels are then incorporated into the bilayer membrane for ion channels activity recordings. The use of Lab-on-a-Chip (LOC) technologies to form lipid bilayer has made lipid bilayer technique efficient for ion channel recordings and has brought great potential for this technique to be scaled up to a high throughput platform for ion channel drug screening. In this review we discuss the use of lipid bilayer for ion channel studies, conventional methods and current LOC technologies on lipid bilayer formations, comparison of lipid bilayer high throughput platform to automated patch clamp system, and finally the potential of lipid bilayer platforms in drug screening assay.

Contents

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1. Introduction

Ion channels are fundamental in many physiological processes, for instance action potential generation, heart and skeletal muscle contractions, transepithelial transports, cell volume regulations, etc. Ion channels are pore structured proteins that allow traverse of specific ions across the lipid bilayer membrane. Ion channels are classified as sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), chloride (Cl^-), and unspecific cation channels based on the types of ions traversable. The selective property of ion channels, together with active pumps and co-transporters generate electrochemical gradient across lipid bilayer membranes. Ion channels undergo cycles of opening and closing, or ‘gating’, to permit ions movement across the channel pore. Regulation of channel gating can be classified as voltage gated, ligand gated, and mechanosensitive gated.

Ion channels are ubiquitously expressed in all biological cell membranes. Malfunction of ion channels can lead to diseases in many tissues such as heart, bone, muscle, kidney and neuron [1,2]. These make ion channels ideal as targets for drug development. About 13% of all marketed drugs target ion channels, and that accounts for $375 million [3]. However, the small numbers of drugs that targets ion channels [4] in contrast to the big amount of money being invested in this market indicates that there is room for improvements in this area.

Several techniques have been employed to study ion channel properties including the lipid bilayer technique. The study of lipid bilayer or artificial cell membrane began in the 1880s (for review see [5,6]). The first artificial lipid bilayer membrane was demonstrated by Mueller et al. in 1962 [7]. The lipid bilayer was formed by ‘painting’ techniques, where lipids in organic solvent were painted over an μm wide hydrophobic aperture, resulting in a ‘free standing’ bilayer. Under reflected light, lipid bilayer appears black and therefore being called Black Lipid Membrane or BLM. In 1969, Bean et al., observed fluctuated conductance of lipid bilayer membrane when bacterial toxin was introduced into the membrane [8]. The discrete in ionic currents were demonstrated later by Ehrenstein et al. [9], and Hladky and Haydon [10] in 1970. To date, it is well known that the fluctuated membrane conductance and ionic current are the result of ion channels or pore forming proteins situated on the lipid bilayer membrane. Based on this concept, lipid bilayer has been extensively used as a platform to study many ion channels [11–19] and pore forming proteins properties [20]. Secondary to ion channel function study, the ability to insert ion channels into lipid bilayer has led to the development of a new biosensor concept, the nanopore sensor, which utilizes the specificity of ion channel function and the change in ionic currents to identify the present/activity of interest biomolecules. Excellent reviews on nanopore sensor have been discussed elsewhere [21–26] and will not be discussed here. In this review we will focus solely on the use of lipid bilayer for ion channel function studies.

Microfabrication techniques are capable of scaling down complicated laboratory procedures into small chips typically referred to as Lab-on-a-Chip, or ‘LOC’ technologies [27]. LOC technologies have emerged as an important aspect in electrophysiology field and in drug discovery investigation [27]. The growing trend in applying LOC technologies in lipid bilayer platforms offers alternative methods for ion channel drug screening assay. These technologies have the potential to entirely change the field of ion channels drug screening in terms of cost and throughput.

In this review, we will discuss the use of lipid bilayer for ion channel study, the conventional technique for lipid bilayer formation, the existing LOC lipid bilayer technologies for lipid bilayer formation and for electrophysiology study, and finally the potential of lipid bilayer in ion channel drug screening assay.

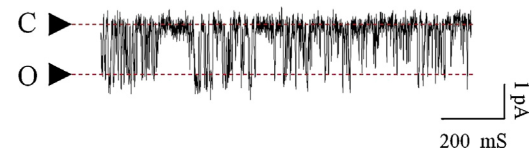

Fig. 1. Example of single channel recording obtained from HEK cells overexpressing small conductance potassium (SK4) channels. Inside out patch clamp configuration was performed under equimolar K^+ concentration on the intra- and extracellular side (140 mM) with 1 μM Ca^{2+} on the intracellular side. The cell membrane patch was clamped at −40 mV. C and O indicate close and open state of the channel respectively.

2. The use of lipid bilayer for ion channel function studies

For the study of ion channel functions, patch clamp is considered the gold-standard technique as it is able to monitor the actions of ion channels in real time, with very high sensitivity and temporal resolution typically in the order of microseconds and sub-picoamperes. Data obtained from patch clamp can be classified as whole cell recordings and single channel recordings. Data obtained from single channel recordings have aided massively in understanding ion channel functions and their role in cell physiology. Single channel recordings in patch clamp are obtained from cell membrane patch. Application of voltage clamp across the membrane causes fluctuations of current which represents the opening and closing state (or gating) of ion channels situated on the cell membrane (Fig. 1). Single channel recordings can also be obtained from a lipid bilayer membrane with ion channels incorporated. The lipid bilayer membrane resembles the cell membrane patch at the tip of the glass capillary in patch clamp. Similar to the patch clamp technique, single channel recordings are achieved by applying voltage clamp across the bilayer membrane in order to monitor changes in ion channel gating.

Lipid bilayer has been used as a model to study the properties of many ion channels for instance acetylcholine receptor [11], sodium channels [12], big conductance potassium channels (BK or Slo1) [19], ryanodine receptors (RyR) [14,16,18], mitochondrial-ATP sensitive potassium channels (mitoK_{ATP}) [17], and purinergic receptors ($P2X_7$) [28]. In many cases lipid bilayer is preferred to patch clamp for example in the study of the lethal ΔF508 Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) mutation. Normal patch clamping of ΔF508 CFTR is not applicable as this mutated protein fails to traffic to the apical membrane and retains in the endoplasmic reticulum (ER) [29]. Therefore, to study them, the purified ΔF508 CFTR channels are being reconstituted onto the lipid bilayer membranes for single channel recordings [30–32]. Moreover, lipid bilayer is being used intensively for the study of intracellular ca^{2+} regulations [16].

3. Conventional techniques for lipid bilayer formation and incorporation of ion channel proteins into lipid bilayer

Lipid bilayer is a well known technique used to study ion channel functions. In conventional technique, lipid bilayer can be formed by three techniques: the ‘painting’ [7], ‘folded’ [33], and ‘vesicle rupture’ techniques. In the painting technique, an experimental chamber is divided into two partitions where both partitions are connected via a small aperture (50–200 μm in diameter) located on a thin hydrophobic septa (in general made of Teflon) (Fig. 2a). After both compartments are filled with aqueous solutions, a lipids/solvent mixture is painted on both sides of the aperture in the two compartments using thin red-sable brush or glass rod to form a monolayer of lipid on each side. The two lipid monolayers are then pulled naturally by van der Waals forces to a close proximity, in a process known as “thining” [7], forming a lipid bilayer. Unlike natural lipid bilayer, the bilayer formed by

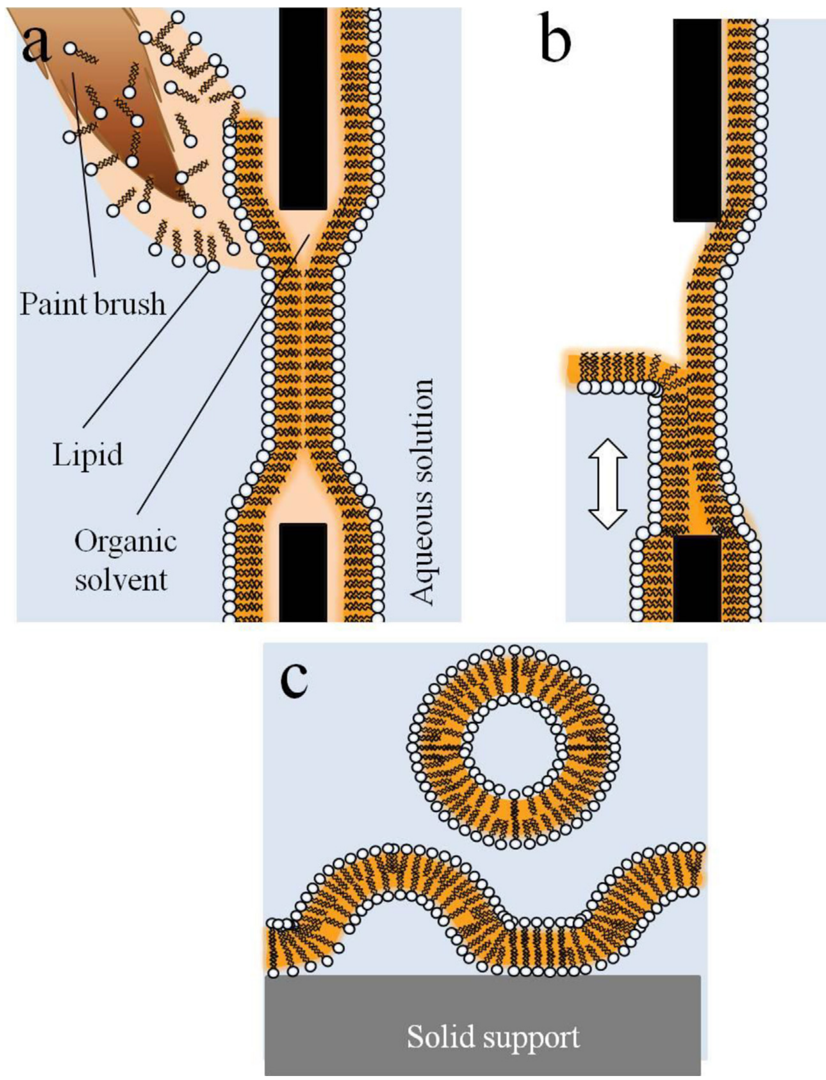

Fig. 2. Lipid bilayer formation techniques. Lipid bilayer can be formed by three standard techniques. (a) 'Painting' technique, lipids/solvent mixture is painted across both sides of the aperture. The lipid monolayers are pulled together to form a lipid bilayer. Lipid solvent is being incorporated into a lipid bilayer membrane, forming annulus of solvent at the edge of the aperture. (b) Folding technique, lipid monolayer is being spread onto water surface, allowing organic solvent to be evaporated. Alteration in water height over the aperture brings the hydrophobic tails of the two monolayers into contact, forming a solvent-less lipid bilayer. (c) Rupture of lipid vesicles, solvent free lipid vesicles are injected onto a solid support surface. Upon contacting the support, lipid vesicles spontaneous rupture and form lipid bilayer.

painting technique incorporates hydrocarbon molecules from the organic solvent thereby affecting lipids thickness. In 1972, Montel and Mueller developed the 'folded technique' to create solvent-less lipid bilayers [33]. In this technique, organic solvent from the lipids/solvent mixture is allowed to evaporate while the lipid monolayer is formed on an air/liquid interface (Fig. 2b). When the height level of the aqueous surface is increased, the monolayer of lipids on the aqueous surface is brought to unite at the aperture, forming a lipid bilayer. Another standard technique is the 'vesicles rupture' technique which is used mainly to create solid supported bilayers. In this technique, a lipid bilayer is formed by depositing solvent free lipid vesicles onto a planar solid support and naturally rupturing them by virtue of the difference in charges between the lipids and the solid surface [34–37] (Fig. 2c). Lipid bilayers formed with this technique are often non-homogenized, for instance the lipid bilayers are not covering the full area of the solid support, resulting in holes in the membranes. In addition, excess deposition of lipid vesicles onto the support causes formation of multilayered lipids.

Many techniques have been established to incorporate ion channel proteins into lipid bilayers. The easiest way to do this is to directly add solubilized proteins in detergent just in front of the lipid bilayer membrane [38]. This method is however harmful to the lipid bilayers as the detergent can rupture the bilayer membranes. A more gentle method is to first fuse the proteins with liposomes to create proteoliposomes that can be directly deposited in front of the prepared membrane. The proteoliposomes will spontaneously fuse to the bilayer membrane. To facilitate proteoliposome fusion into a lipid bilayer, different salt gradients are set up in the two opposite compartments, creating osmotic pressure in the proteoliposomes. The proteoliposomes then rupture, allowing proteins to be inserted into the prepared lipid bilayers [39,40]. Another well known technique is the incorporation of proteins into lipid bilayers from monolayers as described by Schilndler and Rosenbusch [41]. In this technique, proteins are incorporated into lipid vesicles. These vesicles are then spread onto an air/liquid interface, forming a lipid monolayer incorporated proteins. Using the folded technique, the lipid monolayer incorporated proteins is brought to unite with pure lipid monolayer made on the other side of the experimental chamber by raising the buffer level (Fig. 2b). This leads to formation of lipid bilayer incorporated proteins over the aperture. In a planar solid support bilayer, proteins can be first pre-incorporated

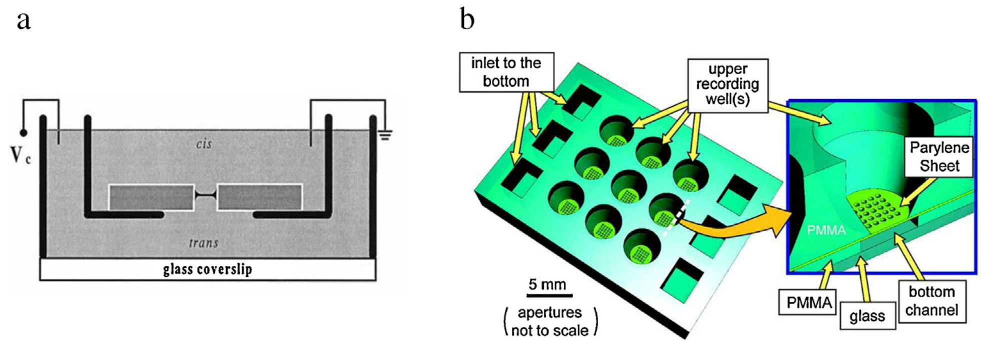

Fig. 3. Micro-aperture devices for free standing lipid bilayer formations. (a) A lipid bilayer formed across the vertical aperture, separating the experimental chamber into two compartments (reprinted with permission from Pantoja et al., Biophysical Journal 81 (2001) 2389–2394). (b) Arrays of micro-apertures for lipid bilayer formations (reprinted with permission from Pioufle et al., Analytical Chemistry 80 (2008) 328–332).

into the lipid vesicles [39]. Deposition of protein incorporated vesicles onto planar solid supports leads to formation of lipid bilayer incorporated proteins.

4. Existing LOC lipid bilayer technologies for ion channel recording

Lipid bilayers formed by conventional techniques are not stable and such operations require skilled labour. Current technologies make use of small scale devices, or the LOC technologies [42] to create platform for lipid bilayer formations which can increase the lipid bilayer stability and facilitate the formation process. In this review, we classify LOC devices for lipid bilayer formation into three major types according to the structure and appearance of the lipid bilayer formed. The three major types are (i) free standing lipid bilayer, (ii) droplet interface bilayer and contacting lipid monolayer, and (iii) supported and tethered lipid bilayer membrane.

4.1. Free standing lipid bilayer membrane

Free standing lipid bilayer represents the closest scenario to the natural environment as the membrane can stand by itself independent of supports at the bilayer region, and both sides of the membrane are exposed to the exchangeable aqueous ionic reservoirs.

Similar to the conventional methods, many LOC lipid bilayer devices make use of a single micro-aperture to form a free standing bilayer. However, unlike the conventional methods, most of the apertures fabricated are vertically orientated (instead of laterally) resulting in formation of lipid bilayers in a horizontal plane (Fig. 3a). The resultant bilayer separates the experimental chamber into upper and lower compartments [43–48]. An alternative to the standalone single aperture is the micro-aperture (30 μm in diameter) array demonstrated by Le Pioufle et al. which has been shown to facilitate formation and increase the stability of the lipid bilayer [46] (Fig. 3b). However, the drawback from this technique is that lipid bilayers must be formed in all micro-apertures in the array or else it cannot be used for ion channel activity measurements.

In general, the apertures are connected to microfluidic channels for exchange of solutions and electrical measurement purposes. Although silicon, due to its well established methods in microfabrication [49], is often used as a material to fabricate the apertures [43–45,50,51], it has been shown that apertures manufactured from parylene sheet [46] as well as other hydrophobic materials including polymethylmethacrylate (PMMA) [52], Teflon [53] and Delrin [48] create a more stable bilayer.

The effect of electrical noise on aperture size was intensively examined by Mayer et al., in which reduction in electrical noise was observed to be associated with smaller apertures fabricated on Teflon [53]. In addition, an increase in the lipid bilayer stability was observed in the relatively smaller apertures. However, in the apertures smaller than 40 μm, noise from additional sources external to the lipid bilayer system becomes dominant.

Various techniques have been demonstrated to form free standing lipid bilayers on micro aperture devices. Conventional painting technique is applied to form lipid bilayers when the apertures are accessible from outside [43,45]. However the success rate and reproducibility in lipid bilayer formation using this technique is low. Ide and Ichikawa have demonstrated a 'streching' technique that speeds up the thinning process thus promoting lipid bilayer formation [54]. The critical step in this technique is that when a bulk of lipids is formed across the aperture, the upper chamber is lifted up slightly to stretch out and thin the lipid bulk to form a lipid bilayer. Another technique to speed up the thinning process is the lipid-air exposure technique as described by Morgan group [52,55]. It works by exposing the lipid bulk at the aperture to air in a repeating cycle. This can be done by intermittently removing and adding back buffer from the upper chamber. In the case when the apertures are externally inaccessible, being located within the hidden geometry of microfluidic channels, it could be done by flowing lipid mixture followed by air into the microfluidic channel [44]. The last technique to create a lipid bilayer is to rupture the lipid vesicles. Sondermann et al. have demonstrated that giant unilamellar vesicles (GUV) are spontaneously ruptured when come into contact with silicon chip surface, resulting in the formation of a free standing lipid bilayer over a silicon aperture [50]. The critical concern for this technique is that the diameter of the vesicles must be larger than the aperture or else the vesicles will slip through the aperture.

The process of incorporating ion channels into lipid bilayers formed on micro-apertures can be performed using conventional techniques such as direct deposition of protein in front of the prepared bilayer membrane or proteoliposome fusion.

4.2. Droplet interface bilayer and contacting lipid monolayer

Droplet interface bilayer or DIB is a technique where a lipid bilayer is created by contacting two aqueous droplets respectively bounded by a lipid monolayer in a lipids/solvent mixture [56–63] (Fig. 4). DIBs can be formed by two different techniques, 'lipid-out' and 'lipid-in' techniques [59,58]. For lipid-out technique, two independent aqueous droplets are being deposited into a lipid-solvent mixture (Fig. 4a) whereas for lipid-in technique, aqueous droplets containing lipid vesicles are being deposited into an organic solvent solution (Fig. 4b). In both techniques, the lipid monolayers spontaneously surround the aqueous surfaces in a manner that the lipid polar heads bound to the aqueous surface and the lipid tails pointing out into the organic solvent solution (Fig. 4a and b). A lipid

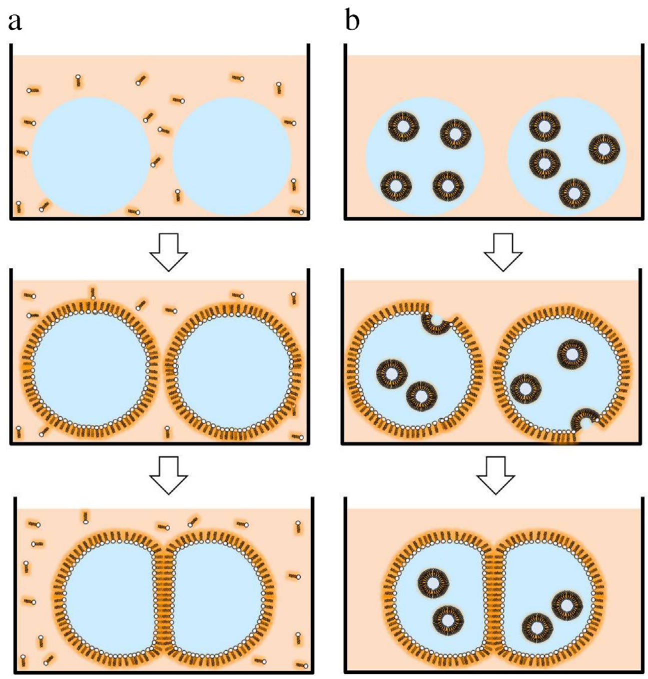

Fig. 4. Droplet interface bilayer (DIB). Diagram illustrates two different techniques to create DIB, (a) lipid-out technique and (b) lipid-in technique [59,58]. (a, upper pannel) In lipid-out technique, aqueous droplets are being deposited in a lipid-solvent mixture, whereas in (b, upper pannel) lipid-in technique, aqueous droplets containg lipid vesicles are being deposited in a solvent solution. (a and b, middle pannel) Lipid monolayer then form spontaneousely around the aqueous droplets, subsequencely (a and b, lower pannel) a lipid bilayer is created by bringing two lipid monolayer droplets into contact.

bilayer can then be formed by bringing the two monolayer droplets into contact [56,58,59,64] by (i) dropping them directly next to each other [56], (ii) manipulating the electrodes that are attached to the droplets [65], or (iii) using electrowetting on dielectric (EWOD) method [64].

Frequently DIB is being integrated with other techniques to form lipid bilayers. Two independent research groups, Zagnoni et al., and El-Arabi et al., have developed microfluidic devices consisting of upper and lower compartments [48,60] (Fig. 5a). The two compartments are connected via a micro-aperture fabricated from Delrin [48] or PMMA [60]. The lower microfluidic channels are filled with aqueous solution whereas the upper compartments are filled with lipid mixtures. Aqueous droplets (2–3 μl) are dropped from the upper comportment onto the apertures where the interface between the droplet and the solution, each consisting of a lipid monolayer, becomes the bilayer. Jung et al. proposed a method to create storable droplets [63] (Fig. 5b). The system is composed of a lipids/solvent mixture (upper layer) and an aqueous solution (lower layer). Upon dropping of aqueous droplet into the lipids mixture, the whole system is immediately being frozen in −80 °C methanol. After thawing, the aqueous droplet is drawn by gravity to the aqueous interface and form a lipid bilayer. Wallace group developed Droplet on Hydrated Support Bilayer (DHB) where a lipid bilayer is formed at the interface between aqueous droplet and hydrogel support [57,62] (Fig. 5c). In this technique, an aqueous droplet, deposited in a lipid-mixture filled compartment above the hydrogel, sediments to the bottom of the compartment and come into contact with the hydrogel support to create a lipid bilayer.

A bilayer formation from DIB technique is formed upon contacting lipid monolayer. Based on contacting lipid monolayer principle, Funakochi et al. have demonstrated a lipid bilayer formation from two aqueous streams [56] (Fig. 5d). In this technique, two aqueous solutions are being pushed from opposite inlets into microfluidic channels filled with lipid-solvent mixture. The two aqueous streams are pushed towards each other, reducing the thickness of the lipid mixture in between them and forming a lipid bilayer. In a similar fashion, a lipid bilayer formed by contacting lipid monolayer of two aqueous streams was demonstrated by Malmstadt et al. [66]. However, in this case, they utilized the property of PDMS to extract the excess solvent out of microfluidic channel allowing formation of solvent free lipid bilayers [66].

Incorporation of ion channels into DIBs and contacting lipid monolayer can be simply made by applying the proteoliposomes into the aqueous droplets/streams. Upon formation of lipid bilayer, ion channels are spontaneously inserted into the lipid bilayer membrane.

Formation of lipid bilayer using DIB and contacting lipid monolayer techniques are efficient because a lipid bilayer can be readily achieved upon contacting of the two monolayers. A possible drawback of DIB technique is that all aqueous solutions are enclosed within the droplets making it difficult for the solutions to be exchanged. Efforts are being made to solve this problem [67,68], however an efficient way of exchanging solutions in DIB system has not yet been demonstrated. The combination of DIB and microfluidic channels [48,60] seem to gain great advantages over conventional DIBs as lipid bilayers can be easily achieved and the buffer solutions can be readily exchanged simply by flowing solutions through microfluidic channels [48,60].

4.3. Supported and tethered lipid bilayers

In the supported bilayer lipid membrane (sBLM) technique (Fig. 6a), lipid bilayer is immobilized onto a planar solid support such as glass, mica, silicon, or metals [34,36,37,69]. The easiest technique for lipid bilayer formation on solid supports is by rupturing unilamellla vesicles on the supports. In this technique, small, large, or giant unilamella vesicles (SUV, LUV, or GUV) are injected onto a solid support (Fig. 2c). Upon vesicles adsorption, they rupture spontaneously due to the oppositely charged lipids and the support surface and form lipid bilayer on the surface [37]. Ion channel incorporation into sBLM can be done either during unilamellas generation which occurs before lipid bilayer is formed, or injected the channels directly onto the prepared bilayer membrane. The close proximity of a lipid bilayer membrane to the supported material often limits insertions of the large transmembrane proteins to the bilayer membrane. Many attempts have been made to extend this distance by using polymer cushions or hydrophilic spacers to tether the bilayer to the support [34,70,71]. Lipid bilayers created by this method are called tethered bilayer lipid membrane (tBLM) (Fig. 6b). Nonetheless measurements of ion channel function were not possible due to the restriction of free ion movements between the membrane and the support. To overcome this problem, Osman group developed tBLM which employed polar spacer tethering lipid bilayer membrane to a gold electrode substrate, creating a gel-like structured hydrophilic region between the membrane and the gold electrode [72,73]. This hydrophilic region serves as a solid-phase ionic reservoir. Using this technique, Osman group was able to record gramicidin conductance using electrochemical impedance spectroscopy technique and single channel recording was demonstrated later by Andersson et al. [74] The single channel recording of gramicidin from tBLM showed signals similar to recordings from the conventional patch clamp using glass pipette [74]. This suggests that the ionic reservoir did not restrict ion movements across the membrane.

Steinem's group integrated the sBLM technique to the free standing lipid technique by creating a porous aluminium membrane with pore sizes of either 55 or 280 nm [75,76] (Fig. 6c). The porous aluminium membrane was assembled between two Teflon chambers filled with buffer solutions and lipids mixture

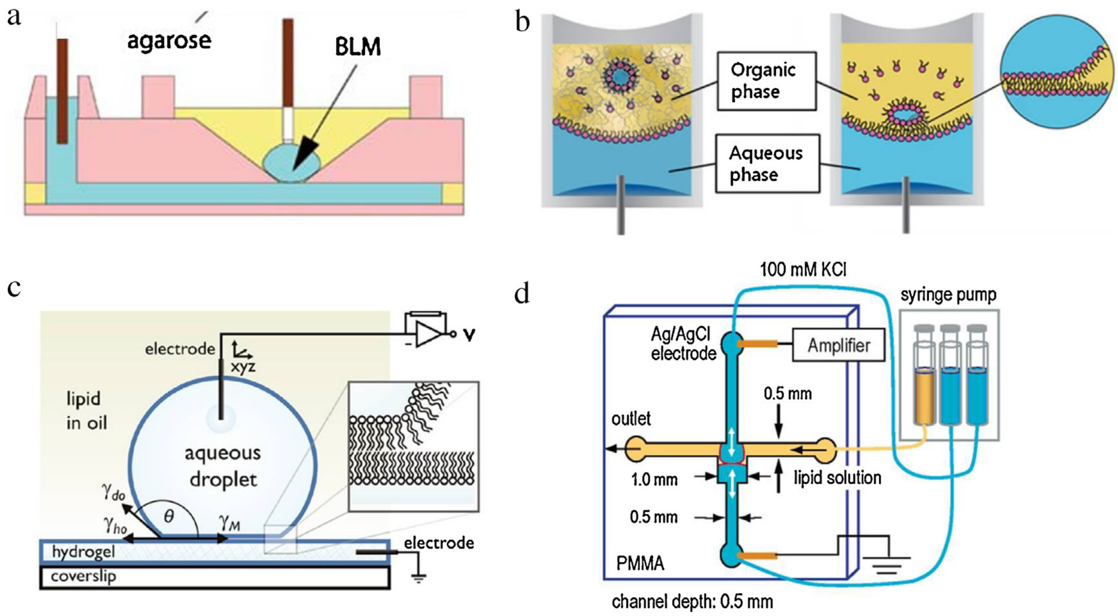

Fig. 5. Example of device uses for DIBs/contacting lipid monolayer formations. (a) A lipid bilayer formed across a micro-aperture by contacting lipid monolayer of an aqueous droplet (upper compartment) and aqueous solution (lower compartment) (reprinted with permission from Zagnoni et al., Analytical and Bioanalytical Chemistry 393 (2009) 1601–1605. Copyright (2009) Springer-Verlag). (b) Storable droplet bilayer system. An aqueous droplet is deposited into a chamber containing lipid mixture solution (upper compartment) and aqueous solution (lower compartment). Immediately after depositing the aqueous droplet into the chamber, the whole chamber is being frozen at −80 °C. A lipid bilayer is formed upon thawing of the chamber by contacting lipid monolayer of an aqueous droplet to an aqueous interface (reprinted with permission from Jung et al., Bioprocess and Biosystems Engineering 35 (2012) 241–246. Copyright (2011) Springer-Verlag). (c) Droplet on hydrogel support. An aqueous droplet is deposited into a lipid mixture solution that covers the hydrogel support. A lipid bilayer is formed above hydrogel support by contacting lipid monolayer of an aqueous droplet and hydrogel interface (reprinted with permission from Gross et al., Langmuir 2011 (27) 14335–14344. Copyright (2011) American Chemical Society). (d) Lipid bilayer formation by contacting lipid monolayer of two aqueous streams. Two aqueous streams are pushed towards each other in cross-shaped microfluidic channels filled with lipid mixture solution, reducing the lipid thickness and forming lipid bilayer (reprinted with permission from Funakoshi et al., Analytical Chemistry 78 (2006) 8169–8174. Copyright (2006) American Chemical Society).

was directly painted onto the *cis* side of the porous membrane. This results in formation of supported lipid bilayers on aluminium substrates and free standing lipid bilayers at the pore sites. Lipid bilayers formation using this method is known as nano-BLM (nBLM).

The benefits of sBLM and tBLM are the robustness and stability of lipid bilayer membranes which can last up to many days. The drawbacks include the possibility of having an unnatural scenario of proteins sitting on or being tethered to the support surface potentially affecting ion channels properties.

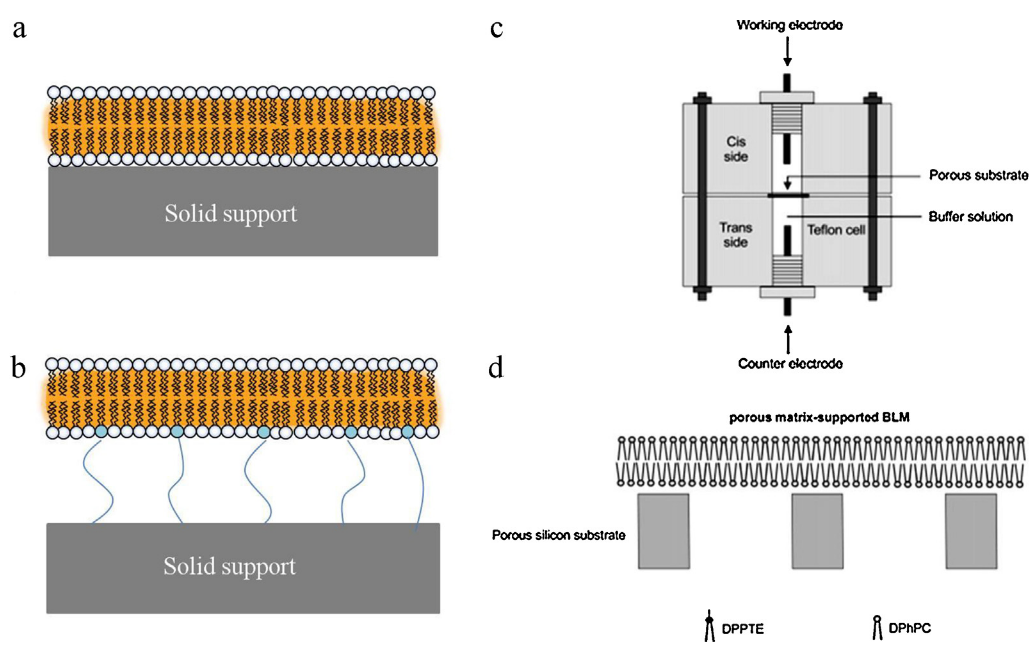

Fig. 6. Illustrations of (a) sBLM, (b) tBLM, and (c) nBLM. (a) In sBLM, a lipid bilayer is formed above a planar solid support, where the bilayer membrane adheres tightly to the solid support. (b) In tBLM, a lipid bilayer is formed above the spacer layer. The spacer tethers the bilayer membrane to the solid support and extends the region between the membrane and the support. (c) In nBLM, a lipid bilayer is formed across porous solid surface, resulting in formation of sBLM above solid support region and free standing lipid bilayer across the porous region (reprinted with permission from Römer et al., Journal of American Chemical Society 126 (2004) 16267–16274. Copyright (2004) American Chemical Society).

Table 1
Prokaryotic/synthetic ion channel recordings from LOC lipid bilayer devices.

Ion channels	Type of lipid bilayer	Techniques used for ion channel activity recordings	Lipid bilayer life time	Material of the aperture (if applicable)	Number of bilayer array (if applicable)	Note	References
Gramicidin	Free standing lipid bilayer	Single channel recordings	15 h	Parylene	9		[46]
	Free standing lipid bilayer	Single channel recordings	15–40 h	Silicon			[51]
	Free standing lipid bilayer	Single channel recordings	2–3 h	PTFE, PMMA	12	50% yeild bilayer formation	[52]
	Free standing lipid bilayer	Single channel recordings		Plastic sheet			[54]
	DIB/contacting lipid monolayer	Single channel recordings	2 h w/o protein insertion 30–60 min w/protein insertion	PMMA			[60]
	DIB/contacting lipid monolayer	Single channel recordings					[61]
	DIB	Single channel recordings	Several hours				[64]
	tBLM	EIS					[72]
	tBLM	EIS					[73]
	nBLM (sBLM + free standing lipid bilayer)	EIS, single channel recordings	Several days	Aluminium			[75]
	Free standing lipid bilayer	Single channel recordings		Teflon			[78]
	Free standing lipid bilayer	Single channel recordings		Silicon	9	60% yeild bilayer formation	[94]
	Free standing lipid bilayer	Single channel recordings		Parylene	96	46% ion channel signals detected	[95]
α-Hemolysin	Free standing lipid bilayer	Single channel recordings		Plastic sheet			[54]
	DIB on hydrogel support	Single channel recordings	72 h-several weeks				[57]
	DIB/contacting lipid mono layer	Single channel recordings	7 h				[62]
	Contacting lipid mono layer	Single channel recordings					[66]
	Free standing lipid bilayer	Single channel recordings	Up to 8 h	Silicon			[77]
	DIB	Single channel recordings			16		[96]
	DIB/contacting lipid monolayer	Single channel recordings	Up to 8 h			Automated system, 81% ion channel signals detected	[79]
	DIB/contacting lipid monolayer	Single channel recordings	More than 1 h		96		[80]
Alamethicin	Free standing lipid bilayer	Single channel recordings			9		[46]
	Free standing lipid bilayer	Single channel recordings					[50]
	Free standing lipid bilayer	Single channel recordings	1–5 h	Teflon			[53]
	DIB/contacting lipid monolayer	Single channel recordings	2 h w/o protein insertion 30–60 min w/protein insertion	PMMA			[60]
Outer membrane protein F (OmpF)	Free standing lipid bilayer	Single channel recordings					[45]
	nBLM (sBLM + free standing lipid bilayer)	EIS, single channel recordings	8–10 h	Aluminium			[76]
Potassium cystallo-graphically sited activation channel (KscA)	DIB/contacting lipid monolayer	Single channel recordings	120 min w/o protein insertion 30–60 min w/protein insertion	PMMA			[60]
	DIB on hydrogel support	Single channel recordings					[91]
Durg discharge outer membrane protein (OprM)	Free standing lipid bilayer	Single channel recordings					[47]
Outer membrane protein G (OmpG)	DIB	Single channel recordings	Several hours-days				[59]
Viral potassium channel (Kcv)	DIB	Single channel recordings			16		[96]
Large conductance mechanosensi-tive channel (MscL)	tBLM	Single channel recordings (using conventional patch clamp)					[82]
Synthetic ion gated channel (SLIC)	tBLM	EIS					[83]

5. Ion channel recordings using lipid bilayer LOC devices

5.1. Data obtained from different types of lipid bilayer

For the study of ion channel activities, single channel recordings can be obtained from free standing bilayers and DIBs [46,51,56,57,60,66,75,77–80] (Table 1). Crucial information that can be extracted from single channel recordings include kinetics of channel gating (close and open lifetime of the ion channel, opening and closing probability), electrical properties of the single ion channel and the membrane patch (single channel conductance, single channel I–V curve, membrane patch

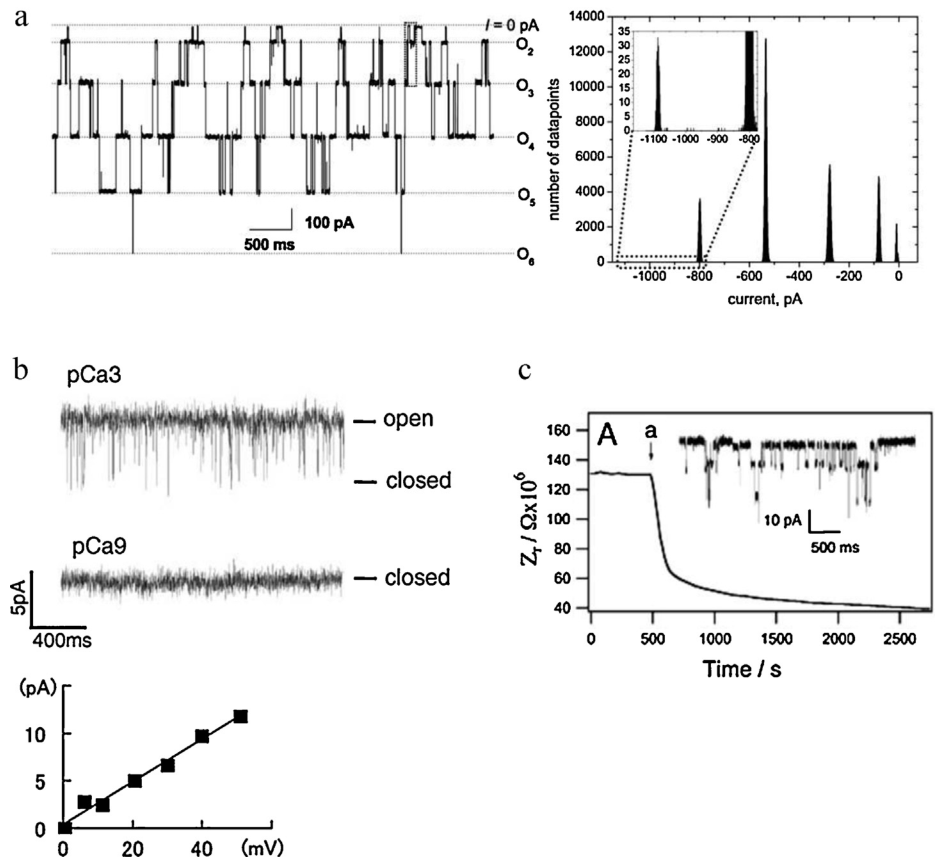

Fig. 7. Example of signal obtained from different types of lipid bilayer. (a) Single channel recordings and peak amplitude obtained from alamethicin reconstituted in free standing lipid bilayer. Lipid bilayer membrane was clamped at -202 mV (reprinted with permission from Mayer et al., Biophysical Journal 85 (2003) 2684–2695. Copyright (2003) Elsevier). (b) Single channel recordings and single channel *I–V* curve obtained from BK channel reconstituted in free standing lipid bilayer. BK channel is active in the present of Ca^{2+} (pCa3) and inactive when Ca^{2+} is withdrawn (pCa9) (reprinted with permission from Ide and Ichikawa, Biosensors and Bioelectronics 21 (2005) 672–677. Copyright (2005) Elsevier). (c) Impedance measurement obtained from SLIC channel reconstituted in tBLM. Decrease in lipid bilayer membrane impedance is observed upon insertion [a] of SLIC into tBLM (reprinted with permission from Terrettaz et al., Langmuir 19 (2003) 5567–5569. Copyright (2003) American Chemical Society).

conductance, and membrane patch *I–V* curve) (Fig. 7a and b) [81].

For tBLM, tethering of lipid bilayer to the conductive solid surface allows direct linking of the membrane to the electrodes [72,73] thereby favouring the system for electrochemical impedance spectroscopy (EIS) technique. Single channel recordings can be obtained from tBLM by subjecting the membrane to conventional patch clamp analysis using glass capillary [74,82,83]. In case of nBLM, both impedance measurements (using EIS) and single channel recordings can be obtained from this type of bilayer membranes [75,76].

EIS is a very powerful technique used intensively in biosensors field [84]. It is normally used to monitor charge transfer from a solution in which an electrode is submerged, to the electrode surface, by measuring impedance. For ion channel recordings, information obtained from EIS includes the distinguish signals of charge transfers on the electrode surface, the ion channels, and the device itself, and the membrane patch conductance (Fig. 7c). However kinetic of ion channel gating cannot be measured because EIS operates at low frequency AC voltage (for review see [85]).

5.2. Ion channels recordings using lipid bilayer LOC devices

Most of the LOC devices developed for lipid bilayer formation use the pore forming antibiotic gramicidin or bacterial toxin α-haemolysin to demonstrate proof-of-concept [46,51,56,57,60,66,75,77–80]. Gramicidin and α-haemolysin are usually preferred because of their well characterized ion channel functions [86–89]. In addition, insertion of these proteins into lipid bilayer membranes is relatively easier as compared to eukaryotic ion channels.

Secondary to the device proof-of-concept, single channel recordings from complex eukaryotic ion channels have been demonstrated on several lipid bilayer LOC devices, particularly on free standing bilayer and DIB systems (Table 2). For instant big conductance potassium channel (BK or Slo1) [43,54], nicotinic receptor [54], ryanodin receptor [54], cold and menthol receptor 1 (CMR1 or TRPM8) [48], voltage gated potassium channel human *Ether-á-go-go* (hERG) [90,91] and many more [91]. These data suggest mature development of lipid bilayer LOC devices for ion channel recordings which can be applied to monitor activity of complicated ion channels.

6. Lipid bilayer in high throughput ion channel recordings

Drug screening process consist of (i) target and hit identification, (ii) hit refinement, (iii) lead refinement, and (iv) regulatory development [92]. For (i) and (ii) effective high throughput systems are required for cost effective and fast screening of compounds in large volume.

6.1. Attempt to develop LOC lipid bilayer technologies for high throughput drug screening assay

Considerable effort has been put to scale up lipid bilayer platform into a higher throughput platform for ion channels recordings, in many cases by application of microfluidic channels into the

Table 2
Eukaryotic ion channel recordings from LOC lipid bilayer devices.

Ion channels	Type of lipid bilayer	Techniques used for ion channel activity recordings	Lipid bilayer life time	Material of the aperture (if applicable)	Number of bilayer array (if applicable)	Note	References
Big conductance potassium channel (BK or Slo1)	Free standing lipid bilayer	Single channel recordings		Silicon			[43]
	Free standing lipid bilayer	Single channel recordings		Plastic sheet			[54]
Human *Ether-á-go-go channel* (hERG)	DIB on hydrogel support	Single channel recordings					[91]
	Free standing lipid bilayer	Single channel recordings		Silicon			[90]
Cold and menthol receptor (TRPM8)	DIB/contacting lipid monolayer	Single channel recordings					[48]
N-methyl-D-aspatate receptor (NMDA-R)	DIB on hydrogel support	Single channel recordings					[91]
Voltage gated potassium channel (Kv1.3)	DIB on hydrogel support	Single channel recordings					[91]
ATP sensitive potassium channel (K_{ATP})	DIB on hydrogel support	Single channel recordings					[91]
Nicotinic receptor	Free standing lipid bilayer	Single channel recordings		Plastic sheet			[54]
Ryanodine receptor	Free standing lipid bilayer	Single channel recordings		Plastic sheet			[54]

systesm [46,52,93]. (i) *Free standing lipid bilayer*. Arays of 9–12 free standing lipid bilayers for simultaneous recordings of ion channel activities have been demonstrated in several studies [46,52,94] (Figs. 3b and 8a). A μm sized aperture for a bilayer formation is fabricated in each measurement unit. Each unit is connected to microfluidic channels for solution loading and exchanging (Fig. 8a). A technique to improve the stability of the free standing bilayer has been proposed by Le Pioufle et al. [46] and Suzuki et al. [95]. A collection of apertures are used to form an array of lipid bilayers for single measurement unit (Figs. 3b and 8b). An increase in membrane stability is claimed to come from a thin parylene sheet used to fabricate small aperture array. (ii) *DIB and contacting lipid monolayer*. DIB arrays were reported by Syeda et al. [96], in which 16-element “DIB-chips” have been proposed for rapid single channel screenings (Fig. 8c). The chip consists of 16 droplets stationed to 16 individual wells connected to Ag/AgCl electrodes and a droplet connected to a movable Ag/AgCl electrode. All droplets are submerged into a lipid-solvent mixture. This results in formation of lipid monolayers around the droplets. A lipid bilayer is formed by contacting the movable monolayer droplet to the each of the stationed droplet. Poulos et al. have presented a 96 well format of DIBs in which a single unit bilayer is formed by contacting monolayer interface of aqueous droplet and aqueous solution [61,80] (Fig. 8d). (iii) *tBLM*. Arrays of 14 tBLMs have been demonstrated by Andersson et al. [82] (Fig. 8e). The gel-like structure was used to form an ionic reservoir region between membrane and the gold electrodes to allow ion movements in tBLM. EIS is the preferred method for ion channel activity measurements on tBLM.

Single channel recordings offer more crucial information as compared to EIS, in particular gating kinetics of ion channels. For this reason, free standing bilayers and DIBs system are preferred to tBLM for ion channel function study and drug screening purpose. To date, only single channel recordings from gramicidin, α-haemolysin, and viral ion channels are demonstrated in the high throughput platform [80,95,96] no eukaryotic ion channels has been reported yet. Demonstration of single channel recordings in eukaryotic ion channels, particularly ion channel-related disease would confers legitimacy to the high throughput lipid bilayer

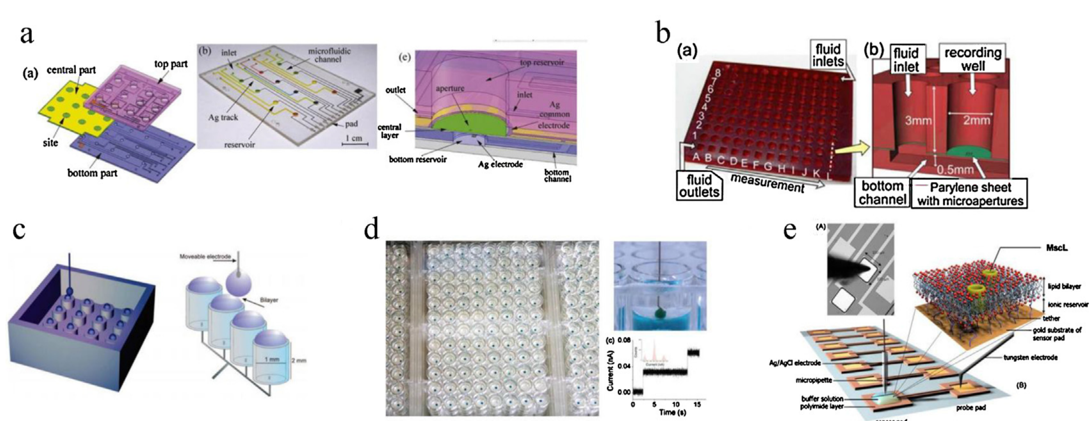

Fig. 8. Example of high throughput LOC lipid bilayer device. (a) Array of 9 free standing lipid bilayer (reprinted with permission from Zagnoni et al., Biosensors and Bioelectronics 24 (2009) 1235–1240. Copyright (2009) Elseveir). (b) 96 well format of free standing lipid bilayer. For each measurement unit, array of micro-aperture are used to form lipid bilayers (reprinted with permission from Suzuki et al., Biomedical Microdevices 11 (2009) 17–22. Copyright (2008) Springer Science + Business Media, LLC). (c) Array of 16 DIBs. 16 aqueous droplets are station at the bottom of micro-chamber chip filled with lipid mixture solution. Lipid bilayer can be formed by bringing the droplet attached to movable electrode into contact with the stationed droplet (reprinted with permission from Syeda et al., Journal of American Chemical Society 130 (2008) 15543–15548. Copyright (2008) American Chemical Society). (d) 96 well format of DIBs. In each measurement unit, lipid bilayer is formed by contacting lipid monolayer of aqueous droplet and aqueous solution (reprinted with permission from Poulos et al., Biosensors and Bioelectronics 24 (2009) 1806–1810. Copyright (2009) Elsevier). (e) Array of 14 tBLM. Lipid bilayers are formed above hydrophilic spacer which tethered the bilayer membrane to gold electrodes (reprinted with permission from Andersson et al., Biosensors and Bioelectronics 23 (2008) 919–923. Copyright (2008) Elsevier).

platform as a tool for ion channel drug screening as well as for ion channel function study.

6.2. Potential of lipid bilayer technique in ion channel drug screening market as compared to automated patch clamp systems

Because of the high quality data obtained from conventional patch clamp, efforts to set up screening throughput based on the patch clamp techniques has led to the development and usage of Automatic Patch Clamp (APC) systems [97–102]. However, the disadvantage of APC system is the relatively high cost-per-data point [100,103]. This gives room to the alternative high throughput platform development which offer high quality data and can produce low cost-per-data point. In this section we intend compare automate patch clamp (APC) system with high throughput lipid bilayer platform. General information regarding both techniques is summarized in Table 3.

6.2.1. Commercial product availability

APC system has been highly cited as a high-quality high throughput platform for ion channels drugs screening [97–102]. Examples of popular high throughput APC systems in the market include Dynaflow HT, IonFlux HT, Ionworks Barracuda, SyncroPatch96, and the QPatch HTX [100,101]. APC developments bring great impact into the ion channel drug screening field because they are automated, provide high quality data, and can perform parallel recordings of up to 384 data points which can greatly shorten the time required for drug screening processes. Recently, APC systems have started to be implemented in laboratories for higher throughput assaying purpose [104,105].

Although ion channel recordings in lipid bilayer technique is well known, array/high throughput LOC device platform is new in the market. Currently there are three commercial products available: (i) the Tethered Membrane System from eDAQ (array of 6), (ii) the bilayer array chip for droplet bilayer formation from Librede (array of 32 and 128), and (iii) the free standing bilayer platform, the Orbit 16, from Nanion (array of 16).

tBLM favours the EIS technique for ion channel activity recordings, however the eDAQ's Tethered Membrane System is able to demonstrate single channel recordings of bacterial machnosensitive channels (MscS/MscK) [106] using 'flying-patch' patch clamp technique [107]. Nevertheless, array of 6 is rather low in throughput, tBLM will not be discussed further in this section. Product from Librede is fabricated based on prove of concept from Schmidt lab [61,79,80], in which a lipid bilayer is formed using DIB technique. The Orbit 16 from Nanion is developed from planar patch clamp chips [50]. A free standing lipid bilayer is formed across a 1 μM wide borosilicate aperture by depositing the giant unilamella vesicle (GUV) onto of the aperture. The GUV then ruptured, forming planar lipid bilayer across the aperture. Although there are commercial products available, application of the array/high throughput lipid bilayer platform in laboratory other than the inventor's own lab has not yet been presented.

6.2.2. In vivo system vs. in vitro system

Data obtained from APC systems come from living cells. Living cells provide data that is closer to physiological environment in which ion channel activities recorded are operated by the natural mechanisms occurring in vivo. However, one of the crucial factors in performing whole cell patch clamp for both conventional and Automatic Patch Clamp is the formation of a gigaseal (a resistance of more than 1 giga-ohm) between the patch clamp substrate and the cell membrane. APC systems, lacking in precise manual control, cannot reliably achieve gigaseal. This would show up as nonspecific current leakage between the membrane and the patch clamp substrate, also known as 'leak current'. The low gigaseal formation rate especially affects APC systems employing population-patch method [102], and would result in drifting baselines.

Data obtained from lipid bilayer platform comes from synthetic bilayer membranes. Researchers have full control over the entire system in vitro. The cost-per-data point can be very cheap as the bilayer platform does not involve the use of living cells for experiments, thus eliminating the cost and time involved in culturing cell lines. The signal channel recorded come solely from the interested channel, as other channels will not be presented on the membrane. Drawbacks of lipid bilayer technique lie in the fact that the system in vitro does not represent the real physiological condition. Missing of some natural components, e.g. protein kinases, enzymes, etc., may affect the interpretation of the experimental results. Another drawback is the low insertion efficiency of ion channels proteins into the bilayer membranes which may result in reduction or absence of the target signals.

6.2.3. Whole cell recordings vs. single channel recordings for ion channel drug screening purposes

The standard procedure to monitor drug effects on ion channel activities is to identify the half maximum inhibitory concentration (IC_{50}) or the half maximum effective concentration (EC_{50}) of the drug. IC_{50} will be evaluated if the compound exerts an inhibitory effect against the ion channel activities whereas EC_{50} will be evaluated if the compound exerts an agonist effect on the ion channel activities. APC technologies use whole cell patch clamp configuration for measuring ion channel activity. Determination of EC_{50} and IC_{50} values of different drugs against different ion channels using APC systems have been demonstrated by Stoelzle et al. [108]. For instance IC_{50} value of erythromycin against hERG is 427.5 μM when the experiments were performed at room temperature and 30.7 μM when the experiments were performed at 35 °C.

High throughput lipid bilayer platform use single channel patch clamp configuration for measuring ion channel activity. Example of single channel recordings used to identify IC_{50} and EC_{50} values has been reported by El-Arabi et al. [48]. Menthol EC_{50} value (111.8 ± 2.4 μM) and 2-APB's IC_{50} value (4.9 ± 0.2 μM) against TRPM8 channel were determined using droplet bilayer chips [48]. In addition to EC50 and IC50 determinations, Syeda et al., have demonstrated screening of different inhibitors against viral potassium channel K_{cv} using droplet bilayer array [96]. The results showed that the entire screening process took less than 1 h for the screening of 9 inhibitors.

Currents obtained from whole cell recordings (APC systems) come from a large population of ion channels. Small changes in the whole cell current can be easily observed because the current recorded is the sum of current from all ion channels. However, because of the large population of ion channels which simultaneously gate, ion channel kinetic is difficult to resolve. On the other hand, currents obtained from single channel recordings (lipid bilayer) come from a small population of ion channels, typically a single channel. Ion channel kinetics can be easily resolved. However, because of the small population of ion channel, the minor changes in ion channel activities are difficult to measure. Despite of all the above, data obtained from both whole cell and single channel recordings are high in quality and suitable for ion channel drug screening.

The discussion above suggests that the high throughput lipid bilayer platform technologies for ion channel recordings still follows behind the APC system. There are many reasons supporting this statement (i) the system does not represent the real physiological condition, (ii) fewer products available in the market, (iii) lower throughput number, (iv) small number of eukaryotic ion channels are studied on high throughput lipid bilayer platform, and (v) no real application in laboratories other than the inventor's. However, some of these reasons stated are not due to the limitation in the

Table 3
Comparison between APC system and high throughput lipid bilayer platform.

Criteria	APC	Array/high throughput system for lipid bilayer platform
Platform	Cell base	In vitro-synthetic bilayer membrane
Information obtained	Whole cell recordings • Whole cell I/V curve • Whole cell conductance	Single channel recording • Channel gating kinetic • Single channel conductance • Membrane patch *I–V* curve
Data quality	High	High
Solution exchangeable	Extracellular and intracellular	Extracellular and intracellular[a]
Cost-per-data point	High [100]	Medium-Low
Throughput	High-up to 1536 well format reported in academic research papers (lateral patch clamp) [97,98] and up to 384 well format commercially available	Medium-up to 96 well format reported in academic research papers and up to array of 128 format commercially available

[a] Eextra- and intracellular solution can be exchanged in free standing lipid bilayer and only extracellular solution can be exchanged in DIB and tBLM.

bilayer system itself but due to lack of the practical performance (e.g. reason number ii–v). Nevertheless, there are many advantages from the high throughput lipid bilayer platform such as the fact that it produces high quality data, is user friendly as lipid bilayer can be readily achieved, confers full control to the researchers over the entire system, records signals solely from interest channels. The two most important advantages relevant for drug screening purpose are that it produces cheaper cost-per-data point and requires shorter time for the whole process as no cell culture is required. Lipid bilayer system is indeed on its rise, one sign indicating this statement is that there is an attempt from APC manufactured firm to perform single channel recordings in lipid bilayer platform using planar patch clamp chips [50].

7. Conclusion

Lipid bilayer is a well known technique that been extensively used for ion channel studies for many decades. For the past decade, applications of LOC technologies for lipid bilayer formation have been widely studied. There are three major types of LOC device for lipid bilayer formation classified by structure and appearance of the lipid bilayer formed (i) free standing bilayers, (ii) DIBs and contacting lipid monolayer, and (iii) sBLMs and tBLMs. Among the three types, free standing bilayer represents the closest environment to natural physiological condition, DIB system seems to be the easiest and most efficient way to form lipid bilayer membrane, and sBLM/tBLM are the most robust bilayer membranes. For ion channel recordings, free standing lipid bilayer and DIB system are more favourable as single channel recordings can be obtained from these two types of bilayer membrane. Array/high throughput platforms of lipid bilayer system for ion channel recordings have been demonstrated, however there is no recordings of eukaryotic ion channels reported.

For ion channel drug screening purpose, high throughput lipid bilayer platform is still following behind APC system in terms of number of product available in the market and throughput. However, lipid bilayer platform has a lot of potential to play a bigger role in ion channel drug screenings. The critical reasons supporting this statement are that lipid bilayer platform provides high-quality data and can produce cheap cost-per-data point. Nevertheless, more research on eukaryotic ion channel recordings and drug screenings are required in lipid bilayer high throughput platform to prove that the system is reliable and able to record signals from complicated ion channel proteins.

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References

[1] C.A. Hubner, T.J. Jentsch, Ion channel diseases, Human Molecular Genetics 11 (2002) 2435–2445.
[2] B. Dworakowska, K. Dołowy, Ion channels-related diseases, Acta Biochimica Polonica 47 (2000) 685–703.
[3] J.J. Clare, Targeting ion channels for drug discovery, Discovery Medicine 9 (2010) 253–260.
[4] J.P. Overington, B. Al-Lazikani, A.L. Hopkins, How many drug targets are there? Nature Reviews Drug Discovery 5 (2006) 993–996.
[5] M. Edidin, Lipids on the frontier: a century of cell-membrane bilayers, Nature Reviews Molecular Cell Biology 4 (2003) 414–418.
[6] M. Adams, Discovering the lipid bilayer, Nature Education 3 (2010) 20.
[7] P. Mueller, D.O. Rudin, H.T. Tien, W.C. Wescott, Reconstitution of cell membrane structure in vitro and its transformation into an excitable system, Nature 194 (1962) 979–980.
[8] R.C. Bean, W.C. Shepherd, H. Chan, J. Eichner, Discrete conductance fluctuations in lipid bilayer protein membranes, The Journal of General Physiology 53 (1969) 741–757.
[9] G. Ehrenstein, H. Lecar, R. Nossal, The nature of the negative resistance in bimolecular lipid membranes containing excitability-inducing material, The Journal of General Physiology 55 (1970) 119–133.
[10] S.B. Hladky, D.A. Haydon, Discreteness of conductance change in biomolecular lipid membranes in the presence of certain antibiotics, Nature 225 (1970) 451–453.
[11] W. Henke, H. Breer, Reconstitution of acetylcholine receptors into planar lipid bilayers, Sub-Cellular Biochemistry 14 (1989) 339–362.
[12] B. Neumcke, Diversity of sodium channels in adult and cultured cells, in oocytes and in lipid bilayers, Reviews of Physiology Biochemistry and Pharmacology 115 (1990) 1–49.
[13] A. Villalobo, Reconstitution of ion-motive transport ATPases in artificial lipid membranes, Biochimica et Biophysica Acta 1017 (1990) 1–48.
[14] M.B. Bhat, S.M. Hayek, J. Zhao, W. Zang, H. Takeshima, W.G. Wier, et al., Expression and functional characterization of the cardiac muscle ryanodine receptor Ca^{2+} release channel in Chinese hamster ovary cells, Biophysical Journal 77 (1999) 808–816.
[15] T.A. Mirzabekov, A.Y. Silberstein, B.L. Kagan, Use of planar lipid bilayer membranes for rapid screening of membrane active compounds, Methods in Enzymology 294 (1999) 661–674.
[16] D.R. Laver, The power of single channel recording and analysis: Its application to ryanodine receptors in lipid bilayers, Proceeings of the Australian Physiological and Pharmaceutical Society 32 (2001) 40–59.
[17] P. Bednarczyk, K. Dołowy, A. Szewczyk, Matrix Mg^{2+} regulates mitochondrial ATP-dependent potassium channel from heart, FEBS Letters 579 (2005) 1625–1632.
[18] A. Marjamaa, P. Laitinen-Forsblom, A. Wronska, L. Toivonen, K. Kontula, H. Swan, Ryanodine receptor (RyR2) mutations in sudden cardiac death: studies in extended pedigrees and phenotypic characterization in vitro, International Journal of Cardiology 147 (2011) 246–252.
[19] J. Liu, A.N. Bukiya, G. Kuntamallappanavar, A.K. Singh, M. Alex, Distinct sensitivity of slo1 channel proteins to ethanol, Molecular Pharmacology 83 (2013) 235–244.
[20] M.W. Parker, S.C. Feil, Pore-forming protein toxins: from structure to function, Progress in Biophysics and Molecular Biology 88 (2005) 91–142.
[21] A. Hirano-iwata, M. Niwano, M. Sugawara, The design of molecular sensing interfaces with lipid-bilayer assemblies, Trends in Analytical Chemistry 27 (2008) 512–520.

[22] M. Saguwara, A. Hirano, P. Bühlmann, Y. Umezawa, Design and application of ion-channel sensors based on biological and artifical receptors, Bulletin of the chemical society of Japan 75 (2002) 187–201.
[23] S. Majd, E.C. Yusko, Y.N. Billeh, M.X. Macrae, J. Yang, M. Mayer, Applications of biological pores in nanomedicine, sensing, and nanoelectronics, Current Opinion in Biotechnology 21 (2010) 439–476.
[24] A. Liu, Q. Zhao, X. Guan, Stochastic nanopore sensors for the detection of terrorist agents: current status and challenges, Analytica Chimica Acta 675 (2010) 106–115.
[25] L. Movileanu, Interrogating single proteins through nanopores: challenges and opportunities, Trends in Biotechnology 27 (2009) 333–341.
[26] L. Gu, J.W. Shim, Single molecule sensing by nanopores and nanopore devices, Analyst 135 (2010) 441–451.
[27] P.S. Dittrich, A. Manz, Lab-on-a-chip: microfluidics in drug discovery, Nature Reviews Drug Discovery 5 (2006) 210–218.
[28] C.A.M. de Souza, P.C.N. Teixeira, R.X. Faria, O. Krylova, P. Pohl, L.A. Alves, A consensus segment in the M2 domain of the hP2X(7) receptor shows ion channel activity in planar lipid bilayers and in biological membranes, Biochimica et Biophysica Acta 1818 (2012) 64–71.
[29] N. Kartner, O. Augustinas, T.J. Jensen, a.L. Naismith, J.R. Riordan, Mislocalization of delta F508 CFTR in cystic fibrosis sweat gland, Nature Genetics 1 (1992) 321–327.
[30] D.N. Sheppard, M.J. Welsh, Structure and function of the CFTR chloride channel, Physiological Reviews 79 (1999) S23–S45.
[31] J. Ma, CFTR chloride channel stimulatory and inhibitory functions of the R domain on CFTR chloride channel, Physiology 15 (2000) 154–158.
[32] D.N. Sheppard, M.A. Gray, X. Gong, Y. Sohma, I. Kogan, D.J. Benos, et al., The patch-clamp and planar lipid bilayer techniques: powerful and versatile tools to investigate the CFTR Cl-channel, Journal of Cystic Fibrosis 3 (2004) 101–108.
[33] M. Montal, P. Mueller, Formation of bimolecular membranes from lipid monolayers and a study of their electrical properties, Proceedings of the National Academy of Sciences of the United States of America 69 (1972) 3561–3566.
[34] E. Sackmann, Supported membranes: scientific and practical application, Science 271 (1996) 43–48.
[35] C. Steinem, A. Janshcff, H. Galla, M. Sieber, Impedance analysis of ion transport through gramicidin channels incorporated in solid supported lipid bilayers, Biochemistry and Bioenergetics 42 (1997) 213–220.
[36] O. Purrucker, H. Hillebrandt, K. Adlkofer, M. Tanaka, Deposition of highly resistive lipid bilayer on silicon–silicon dioxide electrode and incorporation of gramicidin studied by ac impedance spectroscopy, Electrochimica Acta 47 (2001) 791–798.
[37] E.T. Castellana, P.S. Cremer, Solid supported lipid bilayers: from biophysical studies to sensor design, Surface Science Reports 61 (2006) 429–444.
[38] T. Rudel, a Schmid, R. Benz, H. a Kolb, F. Lang, T.F. Meyer, Modulation of Neisseria porin (PorB) by cytosolic ATP/GTP of target cells: parallels between pathogen accommodation and mitochondrial endosymbiosis, Cell 85 (1996) 391–402.
[39] M. Zagnoni, M.E. Sandison, P. Marius, A.G. Lee, H. Morgan, Controlled deliverty of proteins into bilayer lipid membranes on chip, Lab on a Chip 7 (2007) 1176–1183.
[40] H. Repke, A. Bérczi, H. Matthies, Incorporation of brain membrane proteins into planar bilayer lipid membranes, Acta Biologica et Medica Germanica 39 (1980) 657–663.
[41] H. Schindler, J.P. Rosenbusch, Matrix protein from *Escherichia coli* outer membranes forms voltage-controlled channels in lipid bilayers, Proceedings of the National Academy of Sciences of the United States of America 75 (1978) 3751–3755.
[42] D. Mark, S. Haeberle, G. Roth, F. Stetten, R. Zengerle, Microfluidic lab-on-a-chip platforms: requirements, characteristics and applications, Chemical Society Reviews 39 (2010) 1153–1182.
[43] R. Pantoja, D. Sigg, R. Blunck, F. Bezanilla, J.R. Heath, Bilayer reconstitution of voltage-dependent ion channels using a microfabricated silicon chip, Biophysical Journal 81 (2001) 2389–2394.
[44] H. Suzuki, K. Tabata, Y. Kato-yamada, H. Noji, S. Takeuchi, Planar lipid bilayer reconstitution with a micro-fluidic system, Lab on a Chip 4 (2004) 502–505.
[45] S.J. Wilk, L. Petrossian, M. Goryll, T.J. Thornton, S.M. Goodnick, J.M. Tang, et al., Integrated electrodes on a silicon based ion channel measurement platform, Biosensors and Bioelectronics 23 (2007) 183–190.
[46] B. Le Pioufle, H.K. Suzuki, V. Tabata, H. Noji, S. Takeuchi, Lipid bilayer microarray for parallel recording of transmembrane ion currents, Analytical Chemistry 80 (2008) 328–332.
[47] W. Wang, L. Monlezun, M. Picard, P. Benas, O. François, I. Broutin, et al., Activity monitoring of functional OprM using a biomimetic microfluidic device, Analyst 137 (2012) 847–852.
[48] A.M. El-Arabi, C.S. Salazar, J.J. Schmidt, Ion channel drug potency assay with an artificial bilayer chip, Lab on a Chip 12 (2012) 2409–2413.
[49] K.E. Petersen, Silicon as a mechanical material, Proceedings of the IEEE 70 (1982) 420–457.
[50] M. Sondermann, M. George, N. Fertig, J.C. Behrends, High-resolution electrophysiology on a chip: Transient dynamics of alamethicin channel formation, Biochimica et Biophysica Acta 1758 (2006) 545–551.
[51] A. Hirano-Iwata, K. Aoto, A. Oshima, T. Taira, R. Yamaguchi, Y. Kimura, et al., Free-standing lipid bilayers in silicon chips-membrane stabilization based on microfabricated apertures with a nanometer-scale smoothness, Langmuir 26 (2010) 1949–1952.
[52] M. Zagnoni, M.E. Sandison, H. Morgan, Microfluidic array platform for simultaneous lipid bilayer membrane formation, Biosensors and Bioelectronics 24 (2009) 1235–1240.
[53] M. Mayer, J.K. Kriebel, M.T. Tosteson, G.M. Whitesides, Microfabricated teflon membranes for low-noise recordings of ion channels in planar lipid bilayers, Biophysical Journal 85 (2003) 2684–2695.
[54] T. Ide, T. Ichikawa, A novel method for artificial lipid-bilayer formation, Biosensors and Bioelectronics 21 (2005) 672–677.
[55] M.E. Sandison, M. Zagnoni, H. Morgan, Air-exposure technique for the formation of artificial lipid bilayers in microsystems, Langmuir 23 (2007) 8277–8284.
[56] K. Funakoshi, H. Suzuki, S. Takeuchi, Lipid bilayer formation by contacting monolayers in a microfluidic device for membrane protein, Analytical Chemistry 78 (2006) 8169–8174.
[57] A.J. Heron, J.R. Thompson, A.E. Mason, M.I. Wallace, Direct detection of membrane channels from gels using water-in-oil droplet bilayers, Journal of the American Chemical Society 129 (2007) 16042–16047.
[58] H. Bayley, B. Cronin, A. Heron, M.A. Holden, W. Hwang, J. Thompson, et al., Droplet interface bilayers, Molecular Biosystems 4 (2008) 1191–1208.
[59] W.L. Hwang, M. Chen, M.A. Holden, H. Bayley, Asymmetric droplet interface bilayers, Journal of the American Chemical Society 130 (2008) 5878–5879.
[60] M. Zagnoni, M.E. Sandison, P. Marius, Bilayer lipid membranes from falling droplets, Analytical and Bioanalytical Chemistry (2009) 1601–1605.
[61] J.L. Poulos, S.A. Portonovo, H. Bang, J.J. Schmidt, Automatable lipid bilayer formation and ion channel measurement using sessile droplets, Journal of Physics: Condensed Matter 22 (2010) 454105.
[62] L.C.M. Gross, A.J. Heron, S.C. Baca, M.I. Wallace, Determining membrane capacitance by dynamic control of droplet interface bilayer area, Langmuir 27 (2011) 14335–14342.
[63] S.-H. Jung, S. Choi, Y.-R. Kim, T. Jeon, Storable droplet interface lipid bilayers for cell-free ion channel studies, Bioprocess and Biosystems Engineering 35 (2012) 241–246.
[64] J.L. Poulos, W.C. Nelson, T.-J. Jeon, C.-J. "CJ" Kim, J.J. Schmidt, Electrowetting on dielectric-based microfluidics for integrated lipid bilayer formation and measurement, Applied Physics Letters 95 (2009) 013706.
[65] M.A. Holden, D. Needham, H. Bayley, Functional bionetworks from nanoliter water droplets, Journal of the American Chemical Society 129 (2007) 8650–8655.
[66] N. Malmstadt, M.A. Nash, R.F. Purnell, J.J. Schmidt, Automated formation of lipid-bilayer membranes in a microfluidic device, Nano Letters 6 (2006) 1961–1965.
[67] S.A. Portonovo, C.S. Salazar, J.J. Schmidt, Solution exchange in sessile droplet lipid bilayers for IC50 measurements of ion channels, Biophysical Journal 102 (2012) 328a.
[68] Y. Tsuji, R. Kawano, T. Osaki, H. Sasaki, K. Kamiya, Solution exchange of droplet contacting lipid bilayer system, in: Micro Electro Mechanical Systems (MEMS), 2012 IEEE 25th International Conference, 2012, pp. 882–885.
[69] C. Steinem, A. Janshoff, W.P. Ulrich, M. Sieber, H.J. Galla, Impedance analysis of supported lipid bilayer membranes: a scrutiny of different preparation techniques, Biochimica et Biophysica Acta 1279 (1996) 169–180.
[70] E. Sackmann, M. Tanaka, Supported membranes on soft polymer cushions: fabrication, characterization and applications, Trends in Biotechnology 18 (2000) 58–64.
[71] C.A. Naumann, O. Prucker, T. Lehmann, J. Rühe, W. Knoll, C.W. Frank, The polymer-supported phospholipid bilayer: tethering as a new approach to substrate-membrane stabilization, Biomacromolecules 3 (2002) 27–35.
[72] G. Krishna, J. Schulte, B.A. Cornell, R.J. Pace, P.D. Osman, Tethered bilayer membranes containing ionic reservoirs: selectivity and conductance, Langmuir 19 (2003) 2294–2305.
[73] G. Krishna, J. Schulte, B.A. Cornell, R. Pace, L. Wieczorek, P.D. Osman, Tethered bilayer membranes containing ionic reservoirs: the interfacial capacitance, Langmuir 17 (2001) 4858–4866.
[74] M. Andersson, H.M. Keizer, C. Zhu, D. Fine, A. Dodabalapur, R.S. Duran, Detection of single ion channel activity on a chip using tethered bilayer membranes, Langmuir 23 (2007) 2924–2927.
[75] W. Römer, C. Steinem, Impedance analysis and single-channel recordings on nano-black lipid membranes based on porous alumina, Biophysical Journal 86 (2004) 955–965.
[76] E.K. Schmitt, M. Vrouenraets, C. Steinem, Channel activity of OmpF monitored in nano-BLMs, Biophysical Journal 91 (2006) 2163–2171.
[77] M.C. Peterman, J.M. Ziebarth, O. Braha, H. Bayley, H.A. Fishman, Ion Channels and lipid bilayer membranes under high potentials using microfabricated apertures, Biomedical Microdevices 4 (2002) 231–236.
[78] M. Kitta, H. Tanaka, T. Kawai, Rapid fabrication of Teflon micropores for artificial lipid bilayer formation, Biosensors and Bioelectronics 25 (2009) 931–934.
[79] T. Thapliyal, J.L. Poulos, J.J. Schmidt, Automated lipid bilayer and ion channel measurement platform, Biosensors and Bioelectronics 26 (2011) 2651–2654.
[80] J.L. Poulos, T. Jeon, R. Damoiseaux, E.J. Gillespie, K.A. Bradley, J.J. Schmidt, Ion channel and toxin measurement using a high throughput lipid membrane platform, Biosensors and Bioelectronics 24 (2009) 1806–1810.
[81] D.R. Laver, The power of single channel recording and analysis: its application to ryanodine receptors in lipid bilayers, Clinical and Experimental Pharmacology and Physiology 28 (2001) 675–686.
[82] M. Andersson, G. Okeyo, D. Wilson, H. Keizer, P. Moe, P. Blount, et al., Voltage-induced gating of the mechanosensitive MscL ion channel reconstituted in

a tethered lipid bilayer membrane, Biosensors and Bioelectronics 23 (2008) 919–923.
[83] S. Terrettaz, M. Mayer, H. Vogel, Highly electrically insulating tethered lipid vilayers for probing the function of ion channel proteins, Langmuir 19 (2003) 5567–5569.
[84] B. Lindholm-Sethson, J. Nyström, M. Malmsten, L. Ringstad, A. Nelson, Electrochemical impedance spectroscopy in label-free biosensor applications: multivariate data analysis for an objective interpretation, Analytical and Bioanalytical Chemistry 398 (2010) 2341–2349.
[85] M. Mayer, S. Terrettaz, L. Giovangrandi, T. Stora, H. Vogel, Functional analysis of ion channels: planar patch clamp and impedance spectroscopy of tethered lipid membranes, in: J. Cooper, A. Cass (Eds.), Biosensors: A Practical Approach, vol. 1, 2nd ed., Oxford University Press, Oxford, 2003, pp. 153–184.
[86] E. Bamberg, P. Lauger, Terprature-dependent properties of gramicidin A channels, Biochimica et Biophysica Acta 367 (1974) 127–133.
[87] B.A. Wallace, Gramicidin channels and pores, Annual Review of Biophysics and Bioengineering 19 (1990) 127–157.
[88] B. Schuster, U.B. Sleytr, Single channel recordings of alpha-hemolysin reconstituted in S-layer-supported lipid bilayers, Bioelectrochemistry 55 (2002) 5–7.
[89] S.Y. Noskov, W. Im, B. Roux, Ion permeation through the alpha-hemolysin channel: theoretical studies based on Brownian dynamics and Poisson–Nernst–Plank electrodiffusion theory, Biophysical Journal 87 (2004) 2299–2309.
[90] A. Hirano-Iwata, T. Nasu, A. Oshima, Y. Kimura, M. Niwano, Lipid bilayer array for simultaneous recording of ion channel activities, Analytical Sciences 28 (2012) 1049.
[91] S. Leptihn, J.R. Thompson, J.C. Ellory, S.J. Tucker, M.I. Wallace, In vitro reconstitution of eukaryotic ion channels using droplet interface bilayers, Journal of the American Chemical Society 133 (2011) 9370–9375.
[92] K.H. Bleicher, H. Böhm, K. Müller, A.I. Alanine, Hit and lead generation: beyond high-throughput screening, Nature Reviews Drug Discovery 2 (2003) 369–378.
[93] J.S. Shim, J. Geng, C.H. Ahn, P. Guo, Formation of lipid bilayers inside microfluidic channel array for monitoring membrane-embedded nanopores of phi29 DNA packaging nanomotor, Biomedical Microdevices 14 (2012) 921–928.
[94] A. Hirano-iwata, T. Nasu, A. Oshima, Y. Kimura, M. Niwano, Lipid bilayer array for simultaneous recording of ion channel activities, Applied Physics Letters 101 (2012) 023702.
[95] H. Suzuki, B. Le Pioufle, S. Takeuhci, Ninety-six-well planar lipid bilayer chip for ion channel recording fabricated by hybrid stereolithography, Biomedical Microdevices 11 (2009) 17–22.
[96] R. Syeda, M.A. Holden, W.L. Hwang, H. Bayley, Screening blockers against a potassium channel with a droplet interface bilayer array, Journal of the American Chemical Society 130 (2008) 15543–15548.
[97] K.C. Tang, J. Reboud, Y.L. Kwok, S.L. Peng, L. Yobas, Lateral patch-clamping in a standard 1536-well microplate format, Lab on a Chip 10 (2010) 1044–1050.
[98] P. Kongsuphol, T. Bansal, K.C. Tang, K.B. Fang, Electrophysiological characterization of ion channels in beta-cells using silicon based lateral patch clamp device, Sensors and Actuators B: Chemical 173 (2012) 462–467.
[99] L. Polonchuk, Toward a new gold standard for early safety: automated temperature-controlled hERG test on the PatchLiner®, Frontiers in Pharmacology 3 (2012) 3.
[100] C. Farre, N. Fertig, HTS techniques for patch clamp-based ion channel screening – advances and economy, Expert Opinion on Drug Discovery 7 (2012) 515–524.
[101] J. Dunlop, M. Bowlby, R. Peri, D. Vasilyev, R. Arias, High-throughput electrophysiology: an emerging paradigm for ion-channel screening and physiology, Nature Reviews Drug Discovery 7 (2008) 358–368.
[102] A. Finkel, A. Wittel, N. Yang, S. Handran, J. Hughes, J. Costantin, Population patch clamp improves data consistency and success rates in the measurement of ionic currents, Journal of Biomolecular Screening 11 (2006) 488–496.
[103] M. Martina, Ion channel screening: advances in technologies and analysis, Fronteir in Pharmacology 3 (2012) 1.
[104] C.J. Milligan, J. Li, P. Sukumar, Y. Majeed, M.L. Dallas, A. English, et al., Robotic multiwell planar patch-clamp for native and primary mammalian cells, Nature Protocols 4 (2009) 224–255.
[105] W. Balansa, R. Islam, F. Fontaine, A.M. Piggott, H. Zhang, T.I. Webb, et al., Ircinialactams: subunit-selective glycine receptor modulators from Australian sponges of the family Irciniidae, Bioorganic and Medicinal Chemistry 18 (2010) 2912–2919.
[106] E. Petrov, P.R. Rohde, B. Cornell, B. Martinac, The protective effect of osmoprotectant TMAO on bacterial mechanosensitive channels of small conductance MscS/MscK under high hydrostatic pressure, Channels 6 (2012) 262–271.
[107] A.G.Y. Macdonald, B. Martinac, Effect of high hydrostatic pressure on the porin OmpC from *Escherichia coli*, FEMS Microbiology Letters 173 (1999) 327–334.
[108] S. Stoelzel, A. Obergrussberger, A. Brüggemann, C. Haarmann, M. George, R. Kettenhofen, et al., State-of-the-art automated patch clamp devices: heat activation, action potentials, and high throughput in ion channel screening, Fronteirs in Pharmacology 2 (2011) 1–11.

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